

Cycling weather resilience in Flanders

MDA cycling weather project

Jane Shadrina

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Abstract

This report studies cycling traffic in Flanders using automatic bicycle counts joined with weather observations. The data are converted into a daily site-level panel. The analysis then estimates rainy-day loss, compares ridge regression with random forest prediction, and groups counting sites by their estimated response to rain.

The final daily panel has 207,289 site-days from 149 counting sites, covering 2019-10-01 to 2026-03-01. Rain appears in 59.9 percent of site-days and heavy rain in 24.9 percent. Daily counts are lower on rainy days: the mean is 578.1 cyclists on dry days and 432.6 cyclists on rainy days. The ridge regression uses $\alpha = 0.1$ and reaches test $R^2 = 0.450$ on log daily count. The random forest reaches test $R^2 = 0.634$, with 200 trees and maximum depth 20. The ridge regression wet-and-dry prediction comparison estimates a median site-level rain penalty of 15.60 percent.

The final k-means solution uses three clusters after removing 12 unusual sites. The silhouette score is 0.333. Cluster 1 contains 85 sites and has the highest mean rain penalty, 19.34 percent. Cluster 2 contains 40 sites and has the lowest mean rain penalty, 5.73 percent. Cluster 3 contains 12 sites and combines higher rain exposure with moderate rain penalty, 8.04 percent. The high-loss counters and cluster 1 sites are the most direct cases to compare with route exposure, drainage, surface quality, lighting, and weather-station distance.

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1 Project background and relevance

AWV (Agentschap Wegen en Verkeer, the Flemish roads and traffic agency) publishes automatic bicycle count data for Flanders. These counters are used to follow cycling traffic, evaluate infrastructure use, plan maintenance, and support cycling policy [2]. The European Declaration on Cycling also treats cycling as part of sustainable and accessible transport [1]. I estimate which counting sites lose cycling volume when it rains, and by how much.

Rain is common in the daily panel. Some rain is recorded on 59.9 percent of site-days, and heavy rain is recorded on 24.9 percent. Dry days average 578.1 cyclists per site. Rainy days average 432.6. A percentage rain penalty needs the daily volume next to it: a 10 percent drop at a busy counter can remove more cyclists than a 30 percent drop at a quiet counter.

Earlier cycling studies also report lower counts during precipitation, colder temperatures, wind, and other weather conditions [5, 6, 7]. The size of the weather response differs across locations and trip purposes [8]. I use weather resilience as a site-level measure: a site receives a higher resilience index when the fitted model estimates a smaller rainy-day penalty for that site.

2 Data sources and joining

The cycling data come from AWV automatic counting stations. The original AWV data are organized as three connected tables:

- monthly count files, with `site ID` (counting location identifier), `richting` (direction of travel), `type` (detected traffic type), `van` and `tot` (start and end of the counting interval), and `aantal` (number counted);
- directions metadata, with `site ID` (counting location identifier), `richting` (direction of travel), and `naam` (description of the direction);
- site metadata, with `site ID` (counting location identifier), `long` (longitude), `lat` (latitude), `naam` (site name), `gemeente` (municipality), `interval` (counting interval in minutes), and installation date.

The count rows are linked to the site metadata through `site ID` (counting location identifier) and to the direction metadata through `site ID` plus `richting` (traffic direction). The analysis keeps cycling observations and uses the site table to add coordinates and location information. The result is a timestamped cycling table with count volume, site id, direction information, and WGS84 coordinates.

Weather variables come from the Royal Meteorological Institute of Belgium automatic weather station data [3, 4]. The joined data include precipitation, temperature, wind, humidity, pressure, sunshine duration, and short-wave radiation. The final models use a smaller set from this table: one rain amount variable, one wind variable, one solar-radiation variable, humidity, pressure, and temperature. Each cycling site is matched to the nearest weather station, and the distance to that station is kept in the joined table.

3 Data pipeline

The pipeline starts with the joined 15-minute cycling-weather observations. These observations are aggregated into the daily panel. Exploratory plots, ridge regression, and random forest regression use that panel, except for the 15-minute EDA plots, which use a raw sample. The ridge regression wet-and-dry prediction comparison produces the site resilience table. The clustering uses that site resilience table.

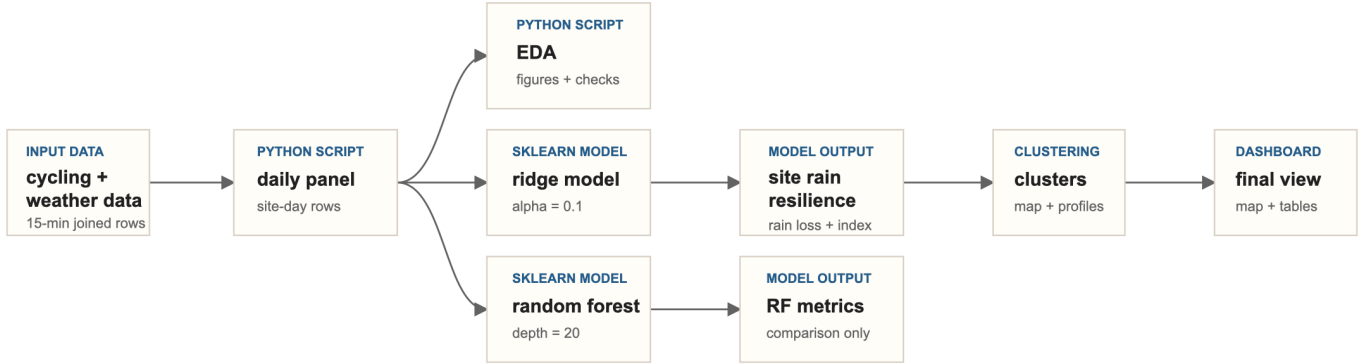


Figure 1: Data transformations and dependencies shown as a directed acyclic graph (DAG).

4 Daily site panel

The joined observations are converted from UTC to Brussels local time. I extract date, hour, weekday, month, and weekend indicators from the local timestamp.

Cycling volume is aggregated to one row per site and day. The daily cyclist count is constructed as the sum of all 15-minute bicycle counts for the same site and Belgian local calendar day. This daily count is the base outcome used later in the supervised models. The daily panel also contains number of observed intervals, commute-hour counts, daily precipitation, rain indicators, daily weather summaries, coordinates, calendar indicators, and distance to the matched weather station.

I fill missing numeric weather and context values with the median inside the modeling pipelines. In the final model table, only commute share has noticeable missingness, at 0.99 percent. The final weather and calendar inputs used in the models have no missing values after the daily panel construction.

The final panel also adds three calendar controls. Belgian public holidays are generated with the Python holidays package using the Belgium calendar. The summer-break flag is coded as July and August. The COVID-period flag is coded from 2020-03-13 to 2021-05-09.

Quantity	Value
Rows	207,289
Sites	149
Date range	2019-10-01 to 2026-03-01
Median daily count	314
Mean daily count	490.9
Site-days with any rain	59.9%
Site-days with heavy rain	24.9%
Median observed slots per site-day	192

Table 1: Daily site panel summary.

5 Exploratory analysis

The temporal plots show whether the expected cycling patterns appear in the data before modeling. Week-day cycling has higher counts around morning and evening commuting hours. Monthly means are lower in winter and higher from spring to early autumn.

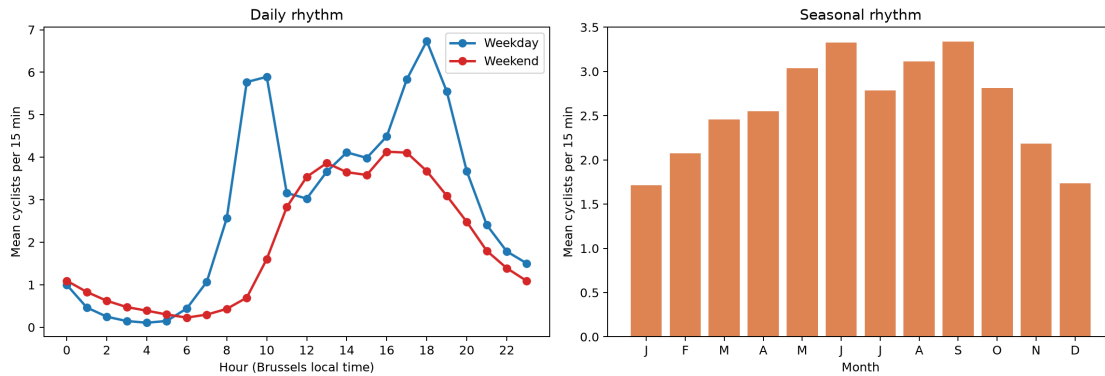


Figure 2: Hourly and monthly cycling patterns.

Missing values are limited in the final modeling table. Commute share is missing in 0.99 percent of site-days. The final weather variables used in the models have no missing values after daily aggregation. The modeling pipelines include median imputation for numeric variables.

Rain is associated with lower observed cycling volume in the raw summaries. Dry days have mean daily count 578.1, while rainy days have mean daily count 432.6. Heavy-rain days have mean daily count 364.9, compared with 532.6 on days without heavy rain. These comparisons include seasonal and site differences, so the model section estimates rain loss with controls.

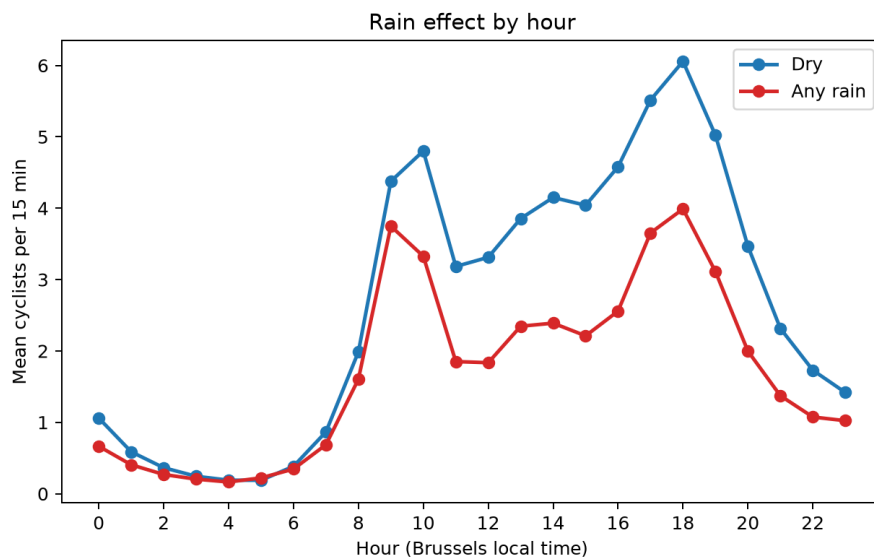


Figure 3: Mean 15-minute cycling counts in dry and rainy intervals.

The site heterogeneity plot shows large differences between counting locations. Some counters have much higher baseline volume, and the wet-vs-dry percentage change also differs across counters. High-volume sites can lose many cyclists even with a moderate percentage drop. Low-volume sites can have a severe percentage drop with a smaller absolute loss.

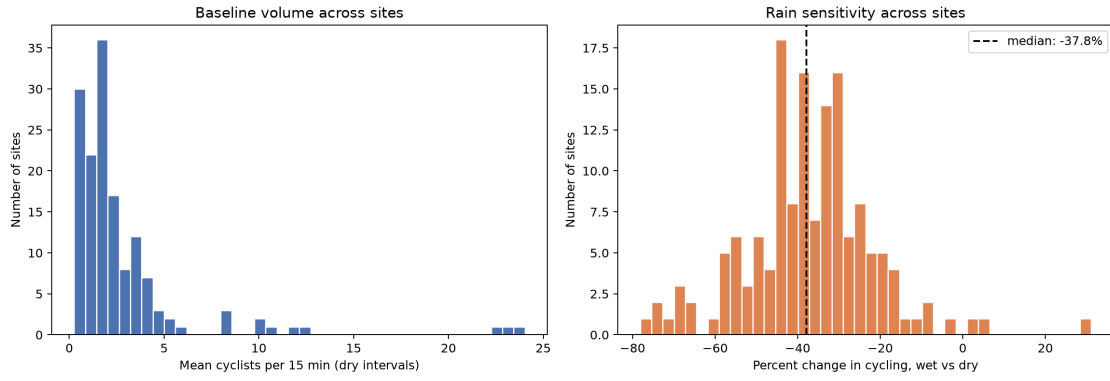


Figure 4: Baseline cycling volume and wet-vs-dry sensitivity across sites.

Weather variables are also correlated with each other. The correlation plot uses log rainfall amount together with the selected daily weather variables. Binary rain indicators are kept in the models as day-type controls: any-rain day and heavy-rain day. The old interval-count rain variables are removed from the final specification. The final model also keeps one wind variable and one solar-radiation variable. The model keeps variables for wet road conditions, thermal comfort, wind exposure, humidity, pressure, and solar conditions.

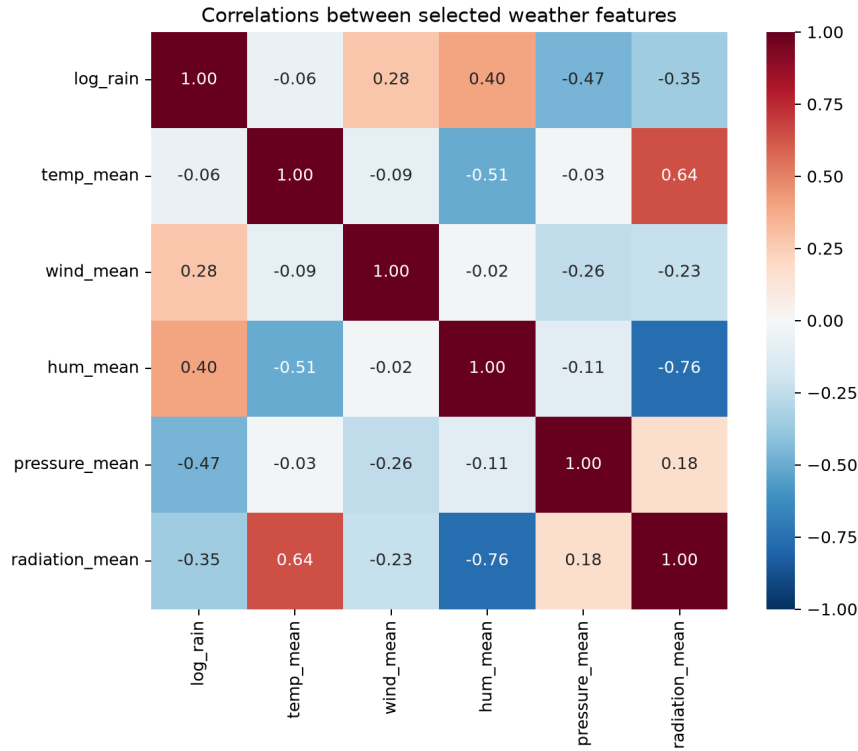


Figure 5: Correlation matrix for the selected daily weather inputs.

6 Ridge regression specification

Ridge regression is used for the weather-resilience calculation. The outcome variable is the log of daily cyclist count. Daily cyclist count means the daily sum of the 15-minute bicycle counts for one site:

$$y = \log(1 + \text{daily cyclist count})$$

The log transformation reduces the dominance of very high-volume counters. A few busy sites would otherwise drive most of the loss function.

Weather features describe rain, temperature, wind, humidity, pressure, and radiation. **Calendar features** include weekend, holiday, month, weekday, and the COVID-period indicator. **Site features** include site identity, commute share, and distance to the matched weather station. **Data coverage features** include the number of observed slots, which controls for incomplete counting days. Highly overlapping rain, wind, and solar variables are kept out of the final specification. The model also includes a **site-specific rainy-day term**, which lets the rain coefficient differ by counter. Ridge regression is used because this specification contains many site-level terms. The ridge penalty shrinks coefficients toward zero and reduces instability. Numeric variables are median-imputed and standardized. Binary variables are imputed with the most frequent value. Categorical variables are one-hot encoded.

The split is chronological: the first 70 percent of dates are used for training, the next 10 percent for validation, and the final 20 percent for testing. The test period starts on 2024-11-18. The data are time-dependent, so random splitting would place nearby dates in both training and test sets.

The tested ridge penalties are $\alpha = 0.1, 1, 10$, and 100. Alpha controls the strength of the L_2 penalty: larger values shrink coefficients more strongly. The best validation RMSE is obtained with $\alpha = 0.1$. The final model is then refit on training plus validation data and evaluated on the held-out test period.

Ridge regression test metric	Value
Selected alpha	0.1
Test start	2024-11-18
RMSE log	0.9611
MAE log	0.4789
R^2 log	0.4502

Table 2: Ridge regression test performance.

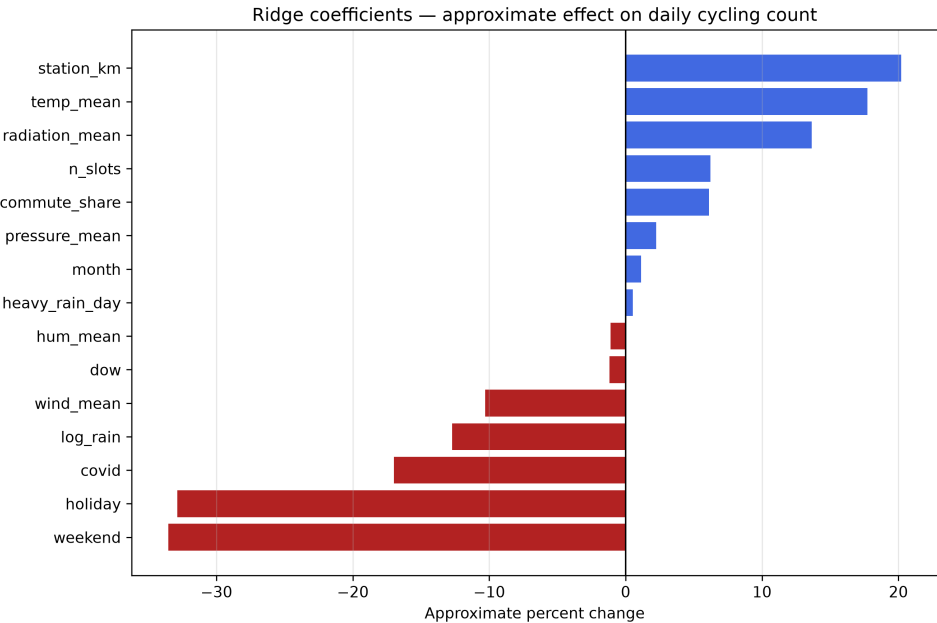


Figure 6: Main ridge regression coefficients as approximate percent changes. Site dummy and site-rain dummy coefficients are removed from the plot.

The largest negative common coefficients are weekend, holiday, the COVID period, log rain, and average wind. The positive temperature coefficient matches earlier cycling-weather findings, where warmer conditions tend to increase cycling volume within normal ranges [6, 7]. The negative rain and wind coefficients are also consistent with previous research [5, 6, 7]. Coefficients are conditional on the other model inputs and on standardization, so they are model associations.

7 Rain penalty and weather resilience

The fitted ridge regression is used to estimate a rain penalty for each site. Rainy rows are predicted twice. First, the model predicts the observed rainy version. Second, the same rows are copied and their rain-related inputs are set to dry conditions. The comparison uses the same site, date, calendar controls, and non-rain weather conditions.

The model predicts $\log(1 + \text{daily cyclist count})$, so the predictions have to be converted back to cyclist counts before calculating a loss in cyclists. The inverse of $\log(1 + x)$ is $\exp(x) - 1$. After this conversion, the estimated rain loss is:

$$\text{rain loss} = \max(\widehat{\text{count}}_{\text{dry}} - \widehat{\text{count}}_{\text{wet}}, 0)$$

The maximum with zero avoids negative loss values when the fitted model predicts a higher count under rain. A zero penalty means that this fitted model did not estimate a positive rain loss for that site under this rule.

For each site, the median loss across rainy days is used as the estimated number of cyclists lost on a rainy day. The median proportional loss is used as the estimated percentage loss. The weather resilience index is:

$$\text{weather resilience index} = 1 - \text{rain penalty percentage}$$

Site-level value	Median	95th percentile	Maximum
Rain penalty percent	15.60%	36.82%	48.48%
Rain penalty count	32.29	152.45	822.28
Weather resilience index	0.844	1.000	1.000

Table 3: Site-level rain penalty and resilience summary.

Site 64 has the largest absolute estimated loss: 822.3 cyclists, with a 43.4 percent rain penalty and median daily count of 1440. Site 48 has the largest percentage penalty: 48.5 percent, with an estimated loss of 283.5 cyclists. Absolute loss ranks busy counters where rainy days remove many cyclists. Percentage loss ranks counters where the relative drop is severe.

8 Random forest specification

The random forest uses the same daily panel, the same outcome variable, and the same simplified weather feature set as the ridge regression. The outcome is again $y = \log(1 + \text{daily cyclist count})$, where daily cyclist count is the summed 15-minute bicycle count for a site-day. The random forest is used for prediction and feature importance. The ridge regression is used for the site-specific wet-and-dry prediction comparison. Numeric variables are median-imputed, binary variables are imputed with the most frequent value, and site identity is one-hot encoded.

The tested settings are 200 or 300 trees, maximum depth 10 or 20, and minimum leaf size 1 or 5. The lowest validation RMSE is obtained with 200 trees, maximum depth 20, and minimum leaf size 1. The random forest has lower held-out error than ridge regression, with test $R^2 = 0.6338$. I use it for prediction quality and feature importance.

Model	RMSE log	MAE log	R^2 log
Ridge regression	0.9611	0.4789	0.4502
Random forest	0.7844	0.4402	0.6338

Table 4: Held-out test performance.

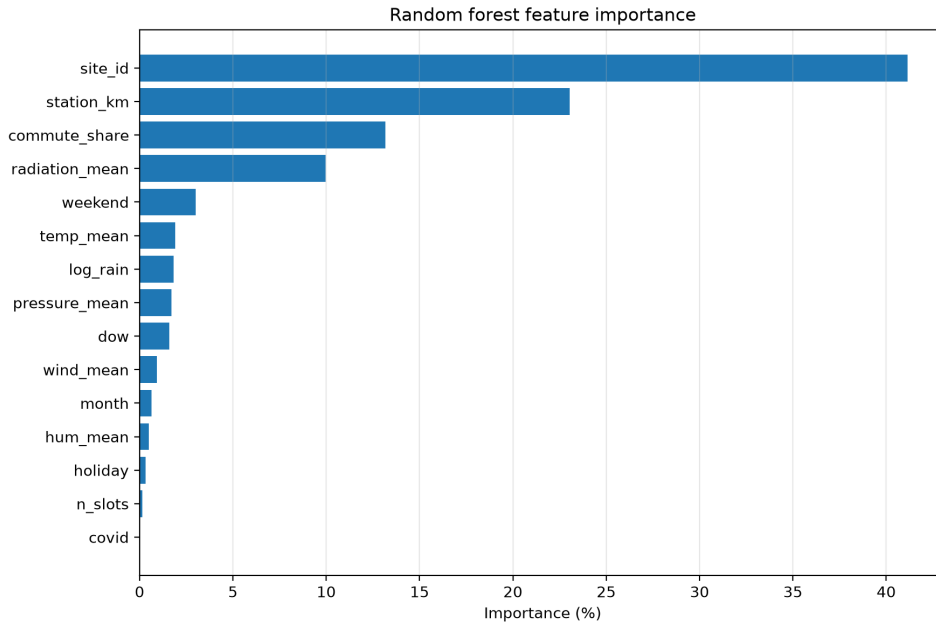


Figure 7: Random forest feature importance.

The largest random forest feature groups are counter identity, weather-station distance, commute share, and short-wave radiation. Counter identity has high feature importance because baseline cycling volume differs strongly by location. The lower random forest error suggests that interactions and nonlinear splits help predict daily count. The ridge regression is used for the rain penalty because its site-specific rainy-day terms match the wet-and-dry prediction comparison.

9 Clustering specification

The clustering uses the site weather-resilience table produced after the ridge regression wet-and-dry prediction comparison. K-means is unsupervised, so the clustering stage has inputs and no outcome variable. Each row is one site, and the inputs are constructed site-level summaries: rain penalty (median predicted percentage/count loss on rainy days), resilience (one minus percentage rain penalty), usual cycling volume (median daily count), rain exposure (share of site-days with rain), commute orientation (median commute-hour share), and distance to the matched weather station.

The variables are first imputed and standardized so that volume, percentages, and station distance are comparable in the clustering. IsolationForest with contamination 0.08 is used before k-means, so 12 out of 149 sites are left outside the final cluster fitting. This threshold was chosen after checking the cluster diagnostics. With 5 percent outliers removed, the $k = 2$ solution already improved. With 8 percent outliers removed, the silhouette score improves further and $k = 3$ becomes the best tested option. PCA with four components is then applied to the scaled variables. The first four principal components retain above 80 percent of the variance, so most of the site-level information stays in the clustering input while correlated variables are compressed into fewer dimensions.

K-means is tested for $k = 2, \dots, 7$, using silhouette score and cluster sizes. The final choice is $k = 3$. Its silhouette score is 0.333, compared with 0.320 for $k = 2$. Three clusters separate a large rain-sensitive group, a stable high-volume group, and a small high-exposure group with moderate rain penalty. The final clusters are:

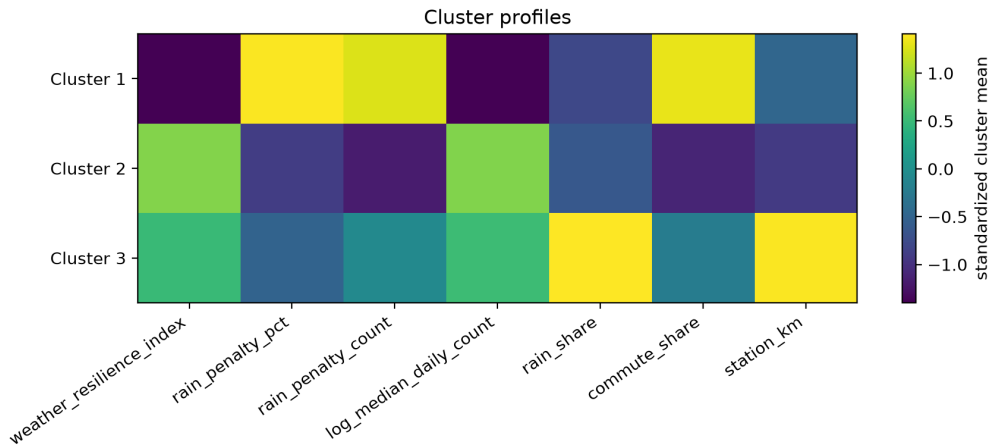


Figure 8: Standardized cluster profiles. Higher values mean the cluster is above the cluster-average level for that variable.

Cluster	Sites	Mean rain penalty	Mean resilience	Mean commute share
1	85	19.34%	0.807	31.3%
2	40	5.73%	0.943	26.8%
3	12	8.04%	0.920	28.4%

Table 5: Final cluster summary. Twelve unusual sites are left without a final k-means cluster after the 8 percent outlier filter.

Cluster 1 contains 85 sites and has the highest mean rain penalty, 19.34 percent. Its mean resilience index is 0.807, the lowest of the three clusters. This is the rain-sensitive group.

Cluster 2 contains 40 sites and has the lowest mean rain penalty, 5.73 percent. Its mean resilience index is 0.943. These sites lose the least estimated cycling volume during rain.

Cluster 3 contains 12 sites. Its mean rain penalty is 8.04 percent. It has high rain exposure, higher usual cycling volume, and longer distance to the matched weather station.

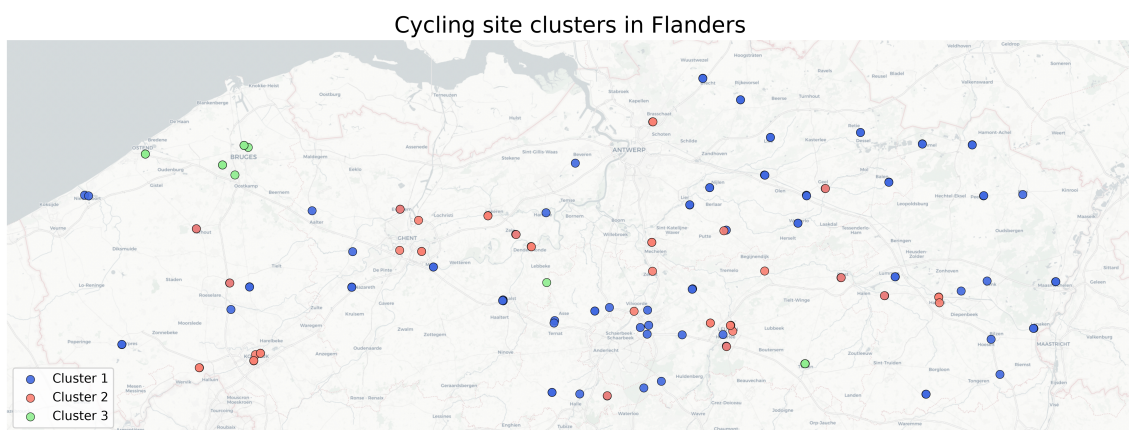


Figure 9: Final site clusters on the Flanders map.

10 Conclusion

Cycling volume is lower on rainy days, and the reduction differs by site. Dry days average 578.1 cyclists and rainy days average 432.6 cyclists. The ridge regression estimates a median site rain penalty of 15.60 percent

by comparing predicted wet and dry versions of the same rainy rows.

The main site-level result is that absolute loss and percentage loss do not rank counters in the same way. Site 64 has the largest absolute estimated loss, 822 cyclists on a rainy day. Site 48 has the largest percentage penalty, 48.5 percent. Site 105 also has a very high percentage penalty, but it is a quiet counter. That is why the report keeps both measures: one shows where many cyclists are lost, the other shows where the relative drop is most severe.

The random forest predicts daily counts better than ridge regression, with test $R^2 = 0.634$. I use ridge regression for the rain-resilience table because it estimates a separate rainy-day effect for each site. That site-level term is then used to compare the same rainy rows under wet and dry settings.

The final clustering has three groups based on the site-level estimates. Cluster 1 is the rain-sensitive group, with 85 sites and mean rain penalty 19.34 percent. Cluster 2 has the lowest estimated penalty, 5.73 percent on average. Cluster 3 is smaller and has higher rain exposure together with higher usual cycling volume.

The final output is a site-level screening tool. It says which counters lose the most cyclists in rain, which counters lose the largest share of their usual volume, and which sites belong to the broader rain-sensitive group. AWW or municipalities could use it before field inspection: start with the high absolute-loss sites when the goal is to protect the largest number of cyclists, and start with high percentage-loss sites when the goal is to find routes that fail most sharply in rain. The cluster labels can also help group similar counters before checking route exposure, drainage, surface quality, lighting, construction history, and weather-station distance.

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